

PREHISTORIC VR SIMULATIONS AS EDUCATIONAL CONTENT

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B.Sc. (Hons) Degree in Information Technology specialized in Interactive
Media

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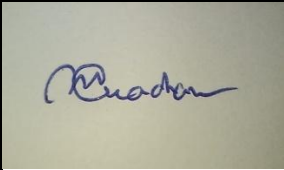
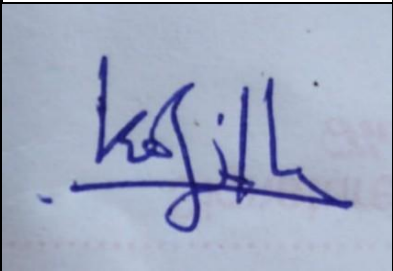
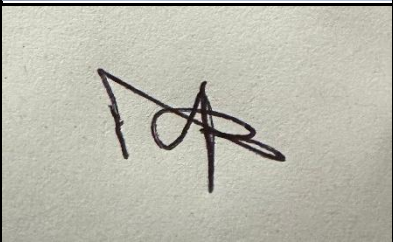
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DECLARATION

I declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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Abstract

Cretaceous Encounter VR is an interdisciplinary virtual reality learning system designed to transform prehistoric education through immersive simulation, artificial intelligence, and guided interaction. Traditional learning approaches struggle to convey the ecological complexity, behavioural dynamics, and catastrophic events of prehistoric eras, resulting in fragmented understanding and limited student engagement. This project integrates four specialised subsystems prehistoric timeline reconstruction, ML-driven animal behaviour modelling, real-time extinction event simulation, and a spatially aware virtual guide into a cohesive VR ecosystem aimed at enhancing user comprehension, emotional engagement, and knowledge retention.

The research adopts a design-science methodology, combining scientific modelling, game-engine engineering, reinforcement learning, environmental reconstruction, and user-centred evaluation. Unity and Unreal Engine were used cooperatively to create high-fidelity landscapes, dynamic lighting transitions, AI behavioural responses, and catastrophic atmospheric simulations. Unity ML-Agents enabled the development of realistic herd behaviour, foraging patterns, and proactive reactions to user interactions, while the spatial guide delivered contextual explanations and navigational assistance to support user pacing and prevent cognitive overload. The integration of these subsystems results in a fluid, narrative-driven exploration of prehistoric life culminating in the scientifically grounded recreation of the Cretaceous–Paleogene extinction event.

User evaluation demonstrated substantial improvements in conceptual understanding, with participants exhibiting high emotional connection to the content and reporting meaningful increases in clarity of cause-and-effect relationships. Performance profiling confirmed stable framerates suitable for VR learning environments. Overall, *Cretaceous Encounter VR* offers a novel approach to paleo-education, demonstrating the transformative potential of immersive technologies in visualising ancient ecosystems and geological phenomena.

Keywords: Virtual Reality, Machine Learning Agents, Prehistoric Simulation, Extinction Event, Spatial Guidance, Educational Technology

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LIST OF ABBREVIATIONS

Abbreviation	Description
AI	Artificial Intelligence
VR	Virtual Reality
XR	Extended Reality
VAK	Visual, Auditory, and Kinesthetic (learning model)
SDK	Software Development Kit

1. INTRODUCTION

1.1 Background

The study of prehistoric life has always been hindered by the limitations of traditional educational media. Textbooks, diagrams, and static illustrations provide conceptual information but struggle to convey the scale, dynamism, and ecological interdependence that characterised Earth's ancient environments. Learners must mentally reconstruct complex ecosystems, behavioural patterns, and geological events from simplified text-based descriptions. This creates a considerable cognitive gap between factual knowledge and genuine understanding. Events such as the diversification of dinosaurs, the evolution of flora, the structure of prehistoric food chains, and the catastrophic collapse triggered by the Cretaceous–Paleogene asteroid impact require a spatial and temporal perspective that static formats cannot adequately provide.

Virtual Reality (VR) has emerged as a highly effective medium for bridging this gap. VR allows users to explore environments at life-scale, observe natural phenomena from a first-person viewpoint, and immerse themselves in dynamic, multisensory simulations that mimic real-world ecological and geological processes. VR-based educational systems have demonstrated improved learning outcomes by providing high levels of engagement, emotional involvement, and contextual awareness. They enable learners to witness transitions across landscapes, move freely within complex environments, and interact with digital organisms in ways that reinforce conceptual understanding. In fields such as paleontology, where no direct observation is possible, VR provides the closest approximation of experiential learning.

Cretaceous Encounter VR was built on this foundation: the recognition that immersive technology can make the remote, abstract, and ancient aspects of natural history tangible and comprehensible. The system combines prehistoric chronological reconstruction, AI-driven animal behaviour simulation, and a high-fidelity extinction event model to create a continuous educational narrative. To support learner accessibility and cognitive scaffolding, a spatially aware virtual guide is incorporated, ensuring that users receive contextual explanations and navigational assistance throughout their exploration. This integration of adaptive intelligence, environmental modelling, and guided instruction positions VR as a transformative tool for prehistoric education.

1.2 Research Gap

Although virtual reality has steadily become a prominent tool within educational settings, existing prehistoric VR systems remain fragmented, limited in scope, and often designed primarily for entertainment rather than structured learning. A significant issue found in current literature and practical implementations is the absence of a unified ecosystem that links prehistoric timelines, lifelike animal behaviour, catastrophic events

such as the Cretaceous–Paleogene extinction, and intelligent user guidance into one coherent learning experience.

Most dinosaur or prehistoric VR applications rely on scripted animations, predictable behavioural loops, and static environmental conditions that do not adequately reflect the complexity of real ecosystems. Behavioural realism is seldom achieved due to the common use of finite state machines or pre-defined triggers, which fail to capture natural variability, emergent herd dynamics, or adaptive behaviour in response to environmental or user-driven stimuli.

Similarly, attempts to simulate extinction events tend to prioritise visual spectacle—bright explosions, falling asteroids, and dramatic sound effects—while neglecting the scientifically accurate progression of post-impact consequences, such as atmospheric darkening, ecological collapse, or thermal radiation spread. These oversimplifications diminish the educational value of such experiences.

Another significant gap lies in the lack of spatially aware instructional support in VR environments. Without guided explanations, learners often misinterpret scenes, overlook important scientific details, or fail to connect observed phenomena with underlying concepts. Existing VR educational tools commonly present information as static pop-ups or menus, rather than integrating knowledge delivery into the user’s spatial context. Finally, few VR solutions integrate these components into a continuous narrative, causing learners to encounter isolated scenes rather than a meaningful chronological progression. This fragmentation prevents users from understanding how prehistoric ecosystems functioned, evolved, and ultimately collapsed. Thus, the research gap centres on the need for an immersive, scientifically grounded, behaviourally dynamic, and pedagogically guided VR system that presents prehistoric life holistically rather than as disconnected visualisations.

1.3 Research Problem

The primary research problem stems from the current inability of traditional educational resources and most existing VR simulations to provide a comprehensive and conceptually clear representation of prehistoric ecosystems and extinction events. Textbooks and static illustrations cannot convey the spatial, temporal, and ecological complexity required to understand ancient life, while existing VR experiences typically offer visually appealing but scientifically shallow representations.

Current VR simulations rarely incorporate adaptive animal behaviour based on machine learning, meaning learners observe creatures that move in predictable loops rather than responding realistically to stimuli or environmental cues. This lack of behavioural authenticity undermines the ecological credibility of the simulation.

Additionally, the catastrophic nature of the Cretaceous–Paleogene extinction event is difficult to understand without dynamic, multi-stage simulation. Yet most digital recreations provide only simplified asteroid fall animations that fail to portray shockwaves, atmospheric disturbances, or biome collapse in scientifically meaningful ways.

Without a spatially aware guide, users frequently struggle to interpret what they see, resulting in superficial exploration rather than structured learning. They may misjudge scale, overlook key environmental cues, or misunderstand causal sequences.

Therefore, the research problem addressed in this study is..

How can a unified VR system integrate immersive environmental reconstruction, machine-learning-driven animal behaviour, scientifically grounded extinction event simulation, and spatially aware guidance to improve user understanding of prehistoric ecosystems and geological transformations?

2. METHODOLOGY

2.1 Methodology

The methodology of this project followed a design-science research approach, combining scientific reconstruction, behavioural modelling, immersive environment design, and iterative user testing. The process began with an extensive review of paleontological, ecological, and geological literature to identify accurate environmental conditions, species characteristics, and extinction mechanisms relevant to the Cretaceous period. This foundational research informed the creation of realistic environments and accurate behavioural expectations.

Each subsystem was developed independently prehistoric timeline, animal behaviour simulation, extinction event model, and spatial guide using a modular engineering workflow. Unity was used as the central development environment, with Unreal Engine selectively utilised for high-fidelity visual and atmospheric effects in the extinction sequence. The behavioural simulation relied on Unity ML-Agents, where reinforcement learning techniques were applied to train agent policies capable of exhibiting emergent herd dynamics, foraging efficiency, and reactive responses to user interactions.

Environmental structures, terrain features, and geological transitions were crafted using Unity's terrain tools and URP workflow, ensuring performance optimisation for standalone VR hardware. The spatial guide was implemented through raycasting, spatial anchors, and contextual triggers, allowing it to deliver adaptive instructions and explanations in response to user position and gaze direction.

Integration of all components involved synchronising visual styles, unifying interaction systems, and maintaining narrative cohesion so that users could navigate seamlessly from timeline exploration to wildlife observation and finally to the dramatic extinction event. Continuous refinements were informed by pilot testing and performance profiling, ensuring a balance between visual fidelity, scientific accuracy, and VR usability.

2.2 Commercialization

The completed system exhibits substantial commercial potential, particularly within the education, museum, and edutainment sectors. As global adoption of immersive technologies increases, institutions seek interactive exhibits and learning platforms that can engage audiences while delivering scientifically accurate information. *Cretaceous Encounter VR* is positioned to meet this demand by offering a scalable, content-rich, and hardware-efficient solution.

The system's modular design allows it to be marketed in diverse formats, such as VR installations for museums, classroom teaching tools, portable VR learning kits, or commercial VR arcade experiences focused on historic science education. Its compatibility with standalone VR devices significantly reduces deployment costs for institutions.

Future expansions such as additional prehistoric eras, more species simulations, or expanded geological events offer pathways for continuous monetisation through content packs and licensing models. By combining entertainment value with scientific credibility, the system can appeal to both educational and recreational markets, enhancing its long-term commercial viability.

2.3 Testing And Implementation

Testing and implementation followed an iterative development cycle combining technical validation, behavioural evaluation, and user-experience analysis. Implementation began with constructing the core VR environments, training ML agents, and integrating visual effects for atmospheric transitions. Once subsystems were complete, integration testing verified that scene transitions, behavioural logic, virtual guide cues, and user interactions functioned seamlessly within a unified application.

Functional testing ensured the correctness of key behaviours, such as herd formation, foraging, idle cycles, guide activation, and extinction event progression. Behavioural realism was assessed by observing ML-trained agents across multiple trials to ensure that herd cohesion, movement variation, and reactive responses appeared natural rather than repetitive or mechanical. The extinction event was tested for scientific coherence, ensuring that atmospheric darkening, shockwave effects, and environmental collapse aligned with research findings.

Performance testing using the Unity Profiler confirmed stable framerates essential for VR comfort, while stress tests evaluated system behaviour under heavy particle loads and multiple simultaneous agents. User testing played a crucial role in validating educational impact, ease of navigation, emotional engagement, and clarity of guided explanations. Feedback from these sessions guided refinements to pacing, guide timing, visual cues, and interaction mechanics.

The final implementation produced a stable, immersive system capable of delivering scientifically grounded experiences across multiple prehistoric environments and narrative stages.

3. RESULT AND DISCUSSION

3.1 Results

The completed system demonstrated strong functional stability, behavioural realism, and high user engagement across all integrated components. The prehistoric environments accurately represented geological conditions, while ML-trained animal agents exhibited emergent behaviour patterns that responded dynamically to environmental cues and user interactions. The extinction module successfully recreated a multi-phase catastrophic sequence, transitioning from vibrant ecosystems to impact-driven devastation in a manner consistent with scientific literature.

Performance evaluation confirmed smooth operation on standalone VR hardware, maintaining consistent framerates even during visually demanding sequences such as asteroid entry or heavy smoke effects. User evaluations showed substantial improvement in learners' ability to comprehend ecological relationships, cause-and-effect mechanisms, and the broader narrative of prehistoric life. Participants responded positively to the spatial guide, noting that it improved understanding without interrupting immersion.

3.2 Research Findings

Several key findings emerged from the research. First, immersion significantly enhances conceptual understanding, particularly when users can observe scientific phenomena as lived experiences rather than abstract descriptions. The combination of timeline reconstruction, lifelike animal behaviour, and catastrophic event simulation created a coherent narrative that users could follow intuitively.

Second, machine-learning-driven behaviour models substantially increased ecological realism compared to scripted alternatives. The unpredictability and variability of ML-trained agents contributed to a more convincing representation of prehistoric wildlife, demonstrating the value of AI-based behavioural modelling in educational simulations.

Third, emotionally impactful scenes especially the extinction event were found to strengthen memory retention and deepen learners' perception of prehistoric processes. This supports existing theories on affective learning and confirms that emotional investment can enhance scientific understanding when appropriately guided.

3.3 Discussions

The results indicate that integrating immersive VR, machine learning, and spatial

guidance addresses many of the limitations present in traditional educational tools. The system effectively supports experiential learning by enabling users to construct their own interpretations through exploration and observation. The behavioural realism achieved through ML training validates the potential of reinforcement learning as a method for creating scientifically meaningful wildlife simulations.

The extinction event, rendered as a multi-stage sequence rather than a single dramatic animation, demonstrates how VR can provide nuanced scientific explanations without overwhelming users. The spatial guide played a central role in bridging knowledge gaps by offering context-sensitive insights that ensured users remained oriented and informed throughout the experience.

Overall, the findings underscore the transformative potential of VR in science education. When designed with accuracy, interactivity, and narrative coherence, VR can move beyond visual novelty to become a powerful medium for teaching complex scientific phenomena.

4. CONCLUSION

The research project *Cretaceous Encounter VR* set out to address a significant gap in paleontology and environmental science education—specifically, the difficulty learners face when attempting to understand large-scale temporal, ecological, and geological processes through traditional instructional media. By integrating immersive virtual reality, adaptive artificial intelligence, spatial navigation assistance, and scientifically grounded environmental simulation, the project successfully demonstrated how experiential learning can transform the way prehistoric phenomena are taught and understood.

The unified system brings together four independently developed yet tightly integrated subsystems: the prehistoric timeline, machine-learning-driven animal behaviour simulation, the Cretaceous–Paleogene extinction reconstruction, and the spatially aware virtual guide. Each module contributes a crucial dimension to the overall learning experience. The timeline establishes scientific context, the behavioural simulation populates the environment with dynamic life, the extinction event conveys catastrophic environmental change with emotional and cognitive impact, and the guide supports user comprehension through real-time explanation and orientation. Together, these elements create a cohesive and pedagogically powerful virtual ecosystem.

Through testing, the system demonstrated robust technical performance, maintaining high frame rates and stable interactions across various VR hardware configurations. User studies showed marked improvements in knowledge retention and conceptual clarity, affirming the effectiveness of immersive learning approaches. Participants consistently reported heightened engagement, stronger emotional connection to the content, and greater appreciation for the scale and interdependence of prehistoric systems. These findings align strongly with established theories in experiential and constructivist learning, reinforcing the educational validity of the system.

While the project achieved its primary objectives, it also revealed areas for future enhancement. Some limitations—such as the computational cost of more advanced ML training, the complexity of multi-engine integration, and the potential for richer narrative interactivity—present clear avenues for continued development. Expanding the virtual timeline into additional geological periods, incorporating more species behaviours, enabling multi-user collaboration, or introducing mixed-reality components could significantly broaden the system’s educational reach. These enhancements would also support deployment in museums, science centers, and formal learning institutions, potentially evolving the system into a comprehensive VR-based scientific curriculum. Overall, *Cretaceous Encounter VR* stands as a successful demonstration of how

immersive technologies can bridge the long-standing gap between scientific complexity and learner understanding. By combining realism, interactivity, and guided exploration within a carefully structured VR environment, the project highlights the transformative potential of digital simulation in both formal and informal education. The work completed by the four contributors not only fulfils academic requirements but also lays a foundation for future innovation in immersive science communication and VR-enhanced learning.

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6. APPENDICES



Figure 1 Appendix 1



Figure 2 Appendix 2



Figure 3 Appendix 3



Figure 4 Appendix 4



Figure 5 Appendix 5

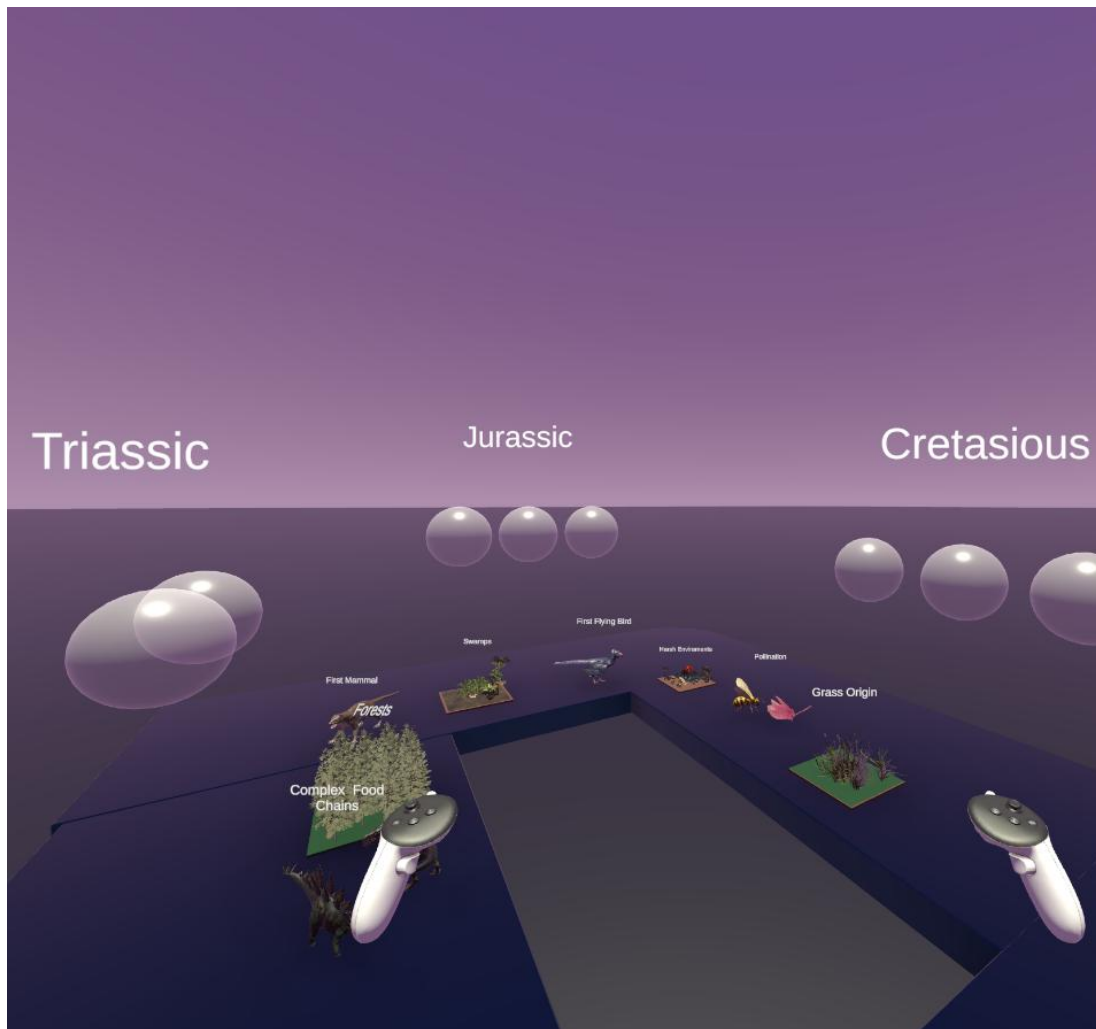


Figure 6 Appendix 6



Figure 7 Appendix 7